According to latest syllabus of
Nepal Engineering Council Registration Examination (NEC)

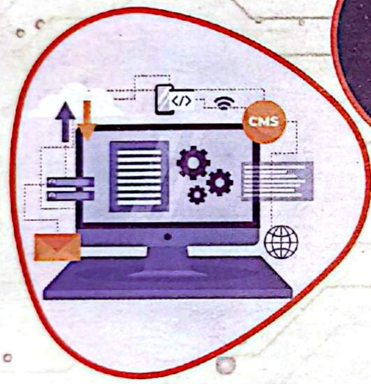
Safal's

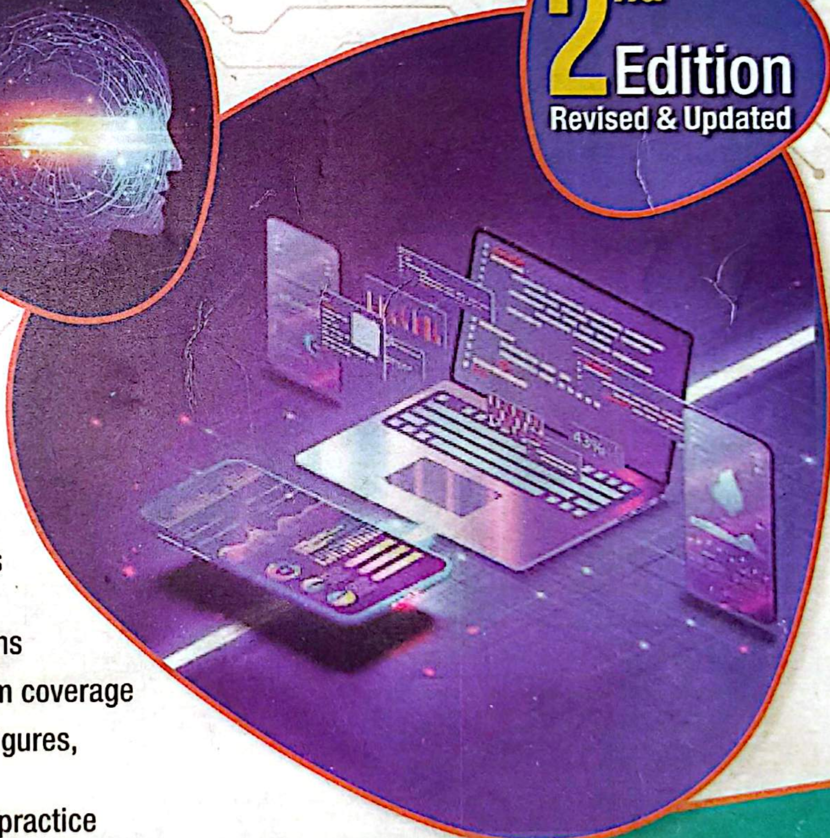
Computer/Software/Information Technology Engineering

Licensure Examinations

COMPREHENSIVE REVIEW

2nd Edition
Revised & Updated





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Er. Suvash Chandra Gautam
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Comprehensive Review

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Preface

The field of **Computer Engineering** is known for being highly competitive, and in order to succeed, aspiring engineers must have a thorough understanding of the subject matter. One critical component of this process is taking **Council Examinations** and other competitive exams. To perform at their best, individuals must be well-prepared. This book is a **Comprehensive MCQ (multiple choice questions)** bank that has been specifically designed to assist aspiring computer engineers in their preparation for these important exams.

The book is designed to cover a wide range of topics in **Computer Engineering, Information Technology Engineering**, and **Software Engineering**. These three disciplines are interconnected and share many common principles and practices, so it is important for students to have a strong understanding of all three.

This book contains over **10,000 MCQs** covering a wide range of sub-disciplines, offering a comprehensive evaluation of the reader's knowledge. The questions have been carefully crafted to reflect the format and level of difficulty of **Council Examinations** and other competitive exams. By practicing with these questions, the reader can familiarize themselves with the types of questions they will likely encounter and gain confidence in their abilities.

Moreover, the book offers detailed explanations of the related theory for each question. This helps the reader to understand not only which answer is correct but also why it is correct. Therefore, the book is an excellent resource for individuals who wish to improve their understanding of **Computer Engineering Syllabus (ACtE), Information Technology Engineering Syllabus (AItE),** and **Software Engineering Syllabus (ASoE)**. It is also a valuable tool for anyone who wants to enhance their exam-taking skills. We have taken great care to create a balanced mix of easy and challenging questions, ensuring that readers of all levels of experience will find the book useful.

Whether you are a student preparing for **Council Examinations** or other competitive exams like **PSC** and **MSc Entrance Examinations**, a practicing engineer seeking to advance your career, or someone simply interested in expanding your knowledge of **Computer Engineering, Information Technology Engineering,** and **Software Engineering,** this book is an essential resource. We are confident that this book will serve as a valuable tool for anyone looking to succeed in the field of Engineering and perform their best in **Council Exams** and **Other Competitive Exams**.

Er. Suvash Chandra Gautam

Er. Ramesh Raj Subedi

Er. Rupak Gyawali

Er. Ganesh Paudel

Er. Roshan Kumar Nandan

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Computer Engineering Syllabus (ACtE)

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3.6 Generic programming and exception handling: Function Template Overloading Functions Template Class Template Function Definition of Class Template. Standard Template Library (Contains, Algorithms, Iterators), Exception Handling Constructs (try, catch throw), Multiple Exception Handling, Rethrowing Exception, Catching All Exceptions, Exception with Arguments, Exceptions Specification for Function Handling Uncaught and Unexpected Exception. (ACtE0306)

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8.2 Software design: Design process, Design Concepts, Design mode Design heuristic, Architectural design decisions, System organization, Modular decomposition styles, Control styles, References architecture, Multiprocessor architecture, Client-server architecture, Distributed object architecture, inter-Organizational distributed computing, Real –time software design and Component-based software engineering. (ACtE0802)

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9.6 Neural Networks: Biological Neural Networks Vs Artificial Neural Network (ANN) McCulloch-Pitts Neuron, Mathematical Model of ANN, Activation functions, Architectures of Neural Networks, The Perceptron, The Learning Rate, Gradient Descent, The Delta Rule, Hebbian Learning, Adaline network, Multilayer perceptron Neural Networks, Backpropagation Algorithm, Hopfield Neural Network. (ACtE0906)

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10.2 Engineering Economics: understanding of project cash flow; discount rate, interest and time value of money; basic methodologies for engineering economics analysis (Discounted Payback Period, NPV, IRR &MARR); Comparison of alternative, depreciation system and taxation system in Nepal. (AALL0102)

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(AALL0103)

10.4 Project Management: Information system; project risk analysis and management; project financing tender and its process, and Contract management.

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